

Supplementary Information A. Hierarchical control architecture

To enable effective use of the reconstructed datasets for traffic control across multiple scales, we provide a modular hierarchical control architecture, as illustrated in Figure S1. The architecture groups control scales into the vehicular, lane-intersection, and network levels, each targeting a distinct control objective. Vehicular-level control regulates individual vehicle maneuvers, lane-intersection-level control governs signal timings and phase operations, and network-level control regulates traffic flow entering the protected regions to prevent oversaturation. Building on the hierarchical traffic state data, we propose a hierarchical control component that incorporates control at multiple scales. The component facilitates the development of new control strategies by extending the hierarchical control structure to leverage the strengths of each control level.

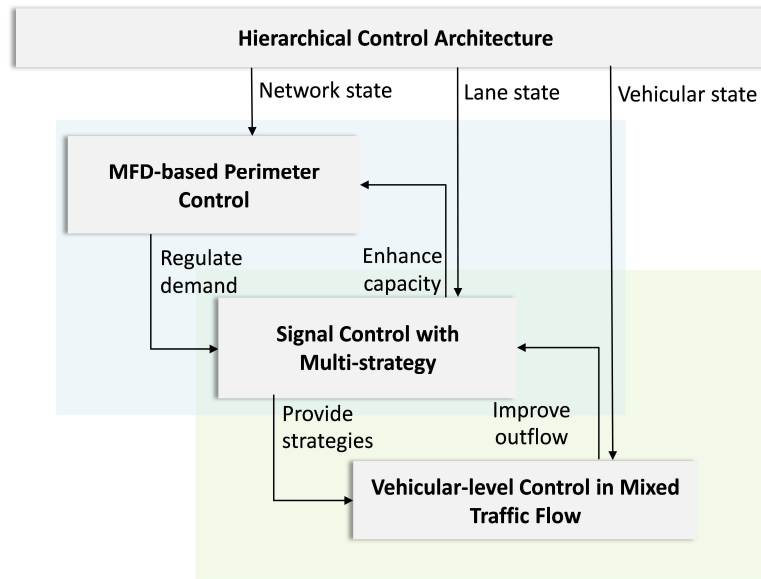


Figure S1. Integrated hierarchical traffic control architecture.

- Vehicular-level control represents the most fundamental level of control, which manages the operation of individual vehicles, including velocity control and routing control. Example applications such as platoon control and vehicle routing, could be conducted as illustrated in Figure S2. In platoon control, the focus lies on longitudinal coordination of connected automated vehicles (CAVs)¹. CAVs are assumed to obtain the state information (e.g., speed, position) of their leading vehicle through vehicular-level perception². They adjust acceleration and deceleration to maintain safe and efficient platoon formations. Velocity control also facilitates more advanced maneuvers, such as platoon merging and splitting³. Vehicle routing determines optimal routes from a set of feasible paths for a given Origin-Destination (OD) pair to minimize travel time⁴.
 - Signal control is one of the most widely used strategies for managing urban traffic at the lane-intersection level⁵. By adjusting signal timings and phase allocations in response to real-time traffic conditions, signal control regulates conflicting traffic streams to optimize intersection throughput, minimize delay, and enhance safety. The control component is therefore designed to support flexible signal control schemes.
- Our study supports the simultaneous deployment of multiple signal control strategies within a network containing heterogeneous intersection layouts, as illustrated in Figure S3. For example, fixed-time control, traffic-responsive control such as max-pressure^{6,7}, and reinforcement learning (RL) based adaptive control can be implemented concurrently. This allows different intersections to operate under control strategies tailored to their specific traffic patterns, thereby improving overall system performance.

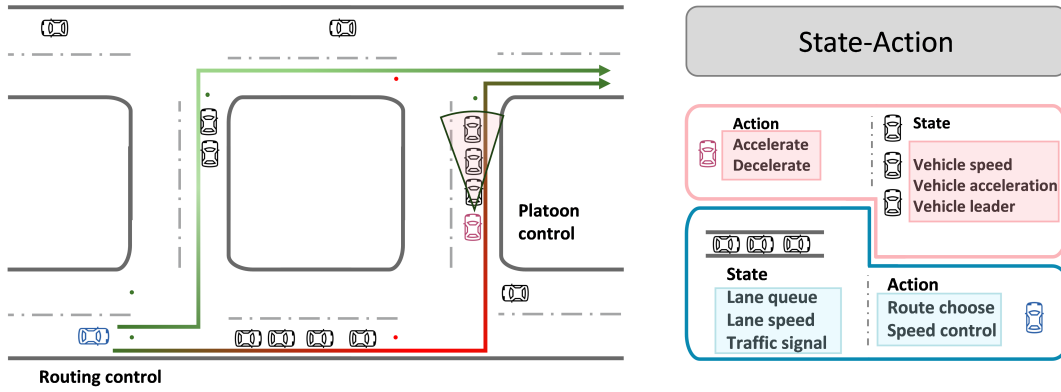


Figure S2. Vehicular-level control comprising platoon control and vehicle routing.

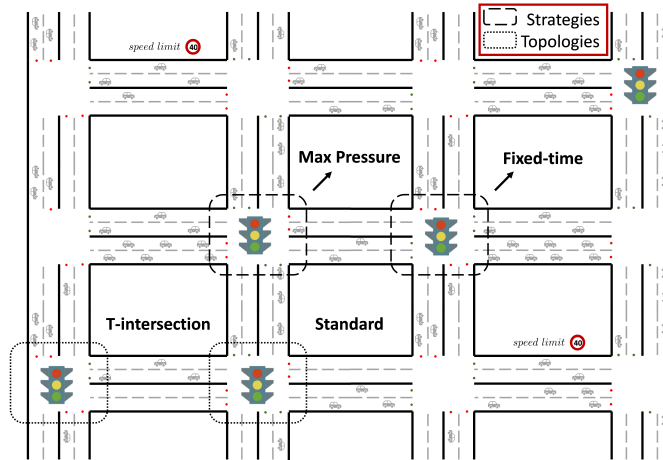


Figure S3. Signal control deployment with different strategies and intersection layouts.

- At the network level, perimeter control has emerged as an efficient strategy for mitigating network-wide congestion while addressing the spatial dimension challenge, facilitated by advances in the macroscopic fundamental diagram (MFD), as illustrated in Figure S4. The MFD describes a unimodal, low-scatter relationship between vehicle accumulation and network production under stationary conditions⁸. Under the MFD framework, a heterogeneous urban network is divided into multiple homogeneous regions with each exhibiting a well-defined MFD⁹, which provides a simplified and straightforward framework for modeling urban network dynamics. The core idea of perimeter control is to regulate the flows entering the protected region (the grey shaded area) at regional boundaries (or critical junctions as red circled) to maximize network throughput¹⁰, thereby enhancing overall network efficiency and stability^{10–16}. This study implements perimeter control at boundary intersections, mapping the transfer flow ratio directly to the green time ratio of the signal. This translates macroscopic flow control into lane-intersection-level operations, facilitating scalable and adaptive control across heterogeneous urban networks.

To conclude, vehicular-level control, lane-intersection-level signal control, and network-level perimeter control interact to support integrated traffic management. For instance, perimeter control determines inflow into a protected network, and it is carried out through signal control¹⁷. Coordinated local signal control regulates the spatial distribution of vehicles within the network, influencing network capacity and the shape of MFDs, and inducing variations in traffic demand patterns. Likewise, coordinated signal control along corridors can be further enhanced by incorporating vehicular-level control, such as speed harmonization and route guidance.



Figure S4. Perimeter control deployed at boundary intersections based on the MFD representation.

Supplementary Information B. Evaluation with multidimensional performance metrics

The metrics are structured according to the hierarchical control levels, as shown in [Table S1](#).

- At the vehicular level, individual vehicle trajectories capture the temporal evolution of vehicle positions within the network. Microscopic traffic metrics are then extracted from these trajectories, including velocity and stop frequency. Velocity is computed as the positional change between consecutive trajectory points divided by the sampling interval. Stop frequency is calculated from vehicle trajectories as the number of times a vehicle's speed falls below a predefined threshold (e.g., 0.1 km/h)¹⁸. By defining analysis zones spanning multiple road segments, these metrics also reflect interactions between vehicles and surrounding traffic entities, such as adjacent vehicles and signal operations.
- At the lane level, we propose a set of mesoscopic indicators, including queue length, density, outflow, average travel time, and average delay. Queue length is calculated as the number of vehicles whose instantaneous speeds fall below the predefined threshold within a lane. Density is calculated as the number of vehicles per unit lane length. Outflow¹⁹ is computed as the number of vehicles exiting the lane per unit time. Average travel time is computed as the mean time required for vehicles to traverse the lane. Average delay quantifies the additional travel time relative to free-flow conditions.
- At the network level, we adopt the macroscopic fundamental diagram (MFD) to characterize the aggregated traffic dynamics of a given network⁸. Three macroscopic performance metrics based on the MFD are induced: (1) The maximum trip completion rate denotes the highest number of trips completed per unit time, reflecting network capacity and the upper bound of traffic efficiency. (2) The critical accumulation corresponds to the vehicle load at which the maximum completion rate is achieved. It indicates the optimal operating state of the network and provides a reference for assessing whether a control strategy can maintain near-peak performance. (3) The MFD-based resilience²⁰ measures the magnitude and duration of reductions in the completion rate when the network deviates from optimal conditions, characterizing the effectiveness and robustness of the network in recovering from congestion and other disruptions.

Supplementary Information C. State data validation with multidimensional evaluation

To ensure rigorous validation, we adopt a multidimensional evaluation scheme in which each control level is assessed using metrics tailored to its objectives. Vehicular-level control is evaluated using vehicle trajectories, while signal control and perimeter control are evaluated using intersection outflow, delay, density distribution, and MFDs.

Table S1. Formulations of multidimensional evaluation metrics across hierarchical control levels.

Level	Metric	Formulation
Vehicular-level	Velocity	$s_n(t) = \frac{p_n(t + \Delta t) - p_n(t)}{\Delta t}$ $p_n(t)$ is the position of vehicle n at time t; Δt is the sampling step.
	Stop frequency	$sf_n = \sum_{t=t_n^{entry}}^{t_n^{exit}} \mathbf{1}(s_n(t) < s_{th})$ s_{th} is a predefined speed threshold; the metric is computed over $[t_n^{entry}, t_n^{exit}]$, when vehicle n is in the analysis zone.
	Queue length	$q_l(t) = \sum_{n \in N_l(t)} \mathbf{1}(s_n(t) < s_{th})$ $N_l(t)$ is the set of vehicles in lane l at time t.
Lane-level	Density	$k_l(t) = \frac{ N_l(t) }{L_l}$ L_l is the length of lane l.
	Outflow	$\Gamma_l(t) = N_l^{exit}(t) $ $N_l^{exit}(t)$ is the set of vehicles leaving lane l over $[t, t + \Delta t]$.
	Average travel time	$TT_l = \frac{1}{ N_l^{exit} } \sum_{n \in N_l^{exit}} T_n$ T_n is the travel time of vehicle n.
	Average delay	$d_l = \frac{1}{ N_l^{exit} } \sum_{n \in N_l^{exit}} (T_n - T_n^*)$ T_n^* is the free-flow travel time of vehicle n.
Network-level	Maximum completion rate	$G_{\max} = \max_a G(a)$ $G(a)$ is the trip completion rate at accumulation a.
	Critical accumulation	$a^* = \arg \max_a G(a)$ a^* is the accumulation at which the trip completion rate reaches its maximum.
	Resilience	$R = - \int_{t_0}^{t_1} (G_{\max} - G(t)) H(a(t) - a^*) dt$ t_0-t_1 is the congestion period; $G(t)$ is the time-varying completion rate; $a(t)$ is the time-varying accumulation; $H(\cdot)$ is the Heaviside indicator.

We conduct experiments to demonstrate the feasibility of developing hierarchical traffic control using hierarchical traffic state data from the Xuancheng dataset. To ensure rigorous validation, we adopt a multidimensional evaluation scheme in which each control level is assessed using metrics tailored to its objectives. Vehicular-level control is evaluated using vehicle trajectories, while signal control and perimeter control are evaluated using intersection outflow, delay, density distribution, and MFDs.

We choose the max-pressure signal control strategy^{6,7}, a benchmark method for managing saturated urban networks, as an example control scenario using the Xuancheng network. Each intersection deploys max-pressure control during the evening peak hour (5:00-6:00 PM). The effectiveness of max-pressure control is assessed using the network MFD (Figure S5, orange points). The trip completion rate increases and reaches a maximum of approximately 6.0 *veh/s* at an accumulation of 5,000 *veh*. Beyond this point, the network becomes congested. At the maximal accumulation of 19,000 *veh*, the completion rate falls below 0.5 *veh/s*, indicating the network is jammed and a substantial decline in overall network performance.

To further investigate the spatial distribution of congestion, lane-level average density is calculated as the temporal mean over the simulation horizon. As shown in Figure S6a, several intersections exhibit higher density levels (red areas). The underlying cause lies in the decentralized and myopic nature of the max-pressure strategy, which prioritizes local pressure gradients and continuously admits inflows from boundary intersections without considering network-wide capacity constraints.

This finding is validated by a lane-level analysis of a representative boundary intersection A, with its location shown in Figure S6a. The analysis focuses on the outflow and delay at this intersection, both aggregated at 60-second intervals. The evolution of its outflow, which corresponds to the inflow from boundary intersection A to the core network, is illustrated in Figure S7a (orange lines). The results indicate that intersection A maintains a high outflow during the early phase of the simulation, even though the internal network is already saturated. After approximately 1,900 seconds, however, spillback leads to gridlock at intersection A, resulting in a sharp decline in outflow to zero. The delay evolution at intersection A is shown in Figure S7b (orange lines). Correspondingly, the intersection maintains low delay before 1,900 seconds, reflecting relatively unconstrained outflow. Once gridlock occurs, delay rises sharply, indicating severe spillback and the onset of congestion propagation. These results indicate that deploying signal control alone is insufficient to prevent network-wide congestion.

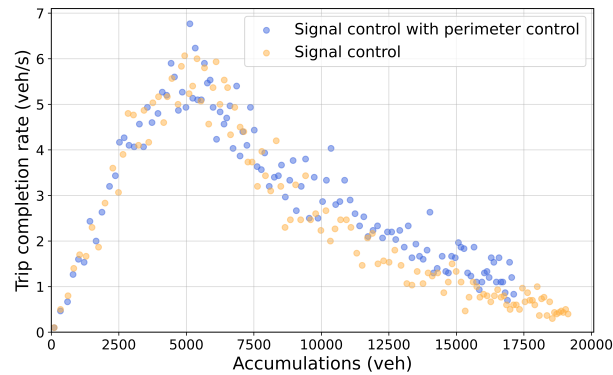


Figure S5. Comparison of MFDs with and without perimeter control.

To remedy this, perimeter control is subsequently introduced to regulate network-level traffic at critical boundary intersections with high inflow demand.¹ The perimeter control strategy implemented in this simulation is empirically designed without fine-tuning, which is beyond the scope of this paper. The controlled intersections are marked by red circles in Figure S4. The resulting MFD (Figure S5, blue points) exhibits a higher maximum completion rate of about 6.8 *veh/s*. Compared with signal-only control, perimeter control improves throughput at medium to high accumulations and constrains network accumulation to below 17,500 *veh*, thereby mitigating oversaturation. The

¹Note that the perimeter control strategy implemented in this simulation is designed empirically without fine-tuning. Fine-tuning the perimeter control strategy is beyond the scope of this paper.

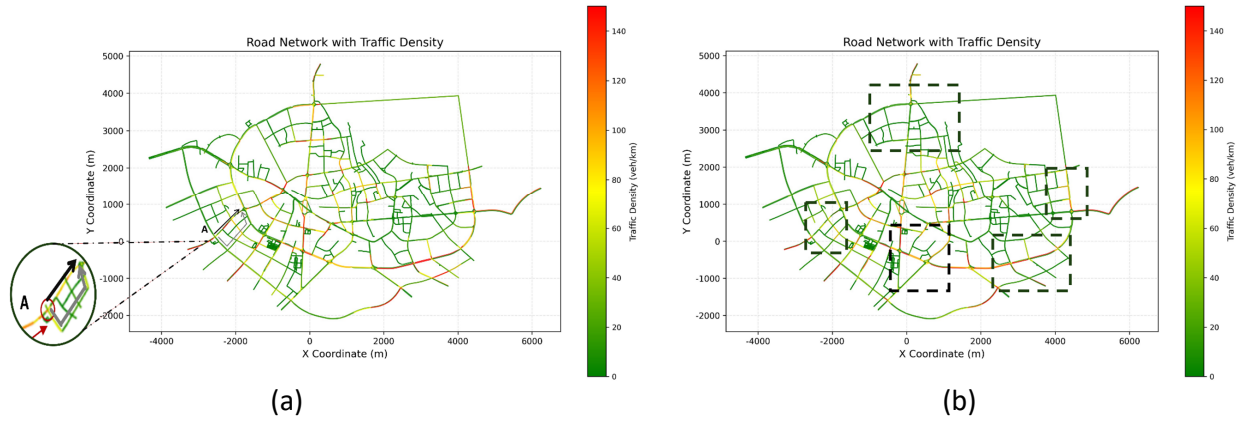


Figure S6. Comparison of traffic density distributions with and without perimeter control. (a) Signal control. (b) Signal control with perimeter control.

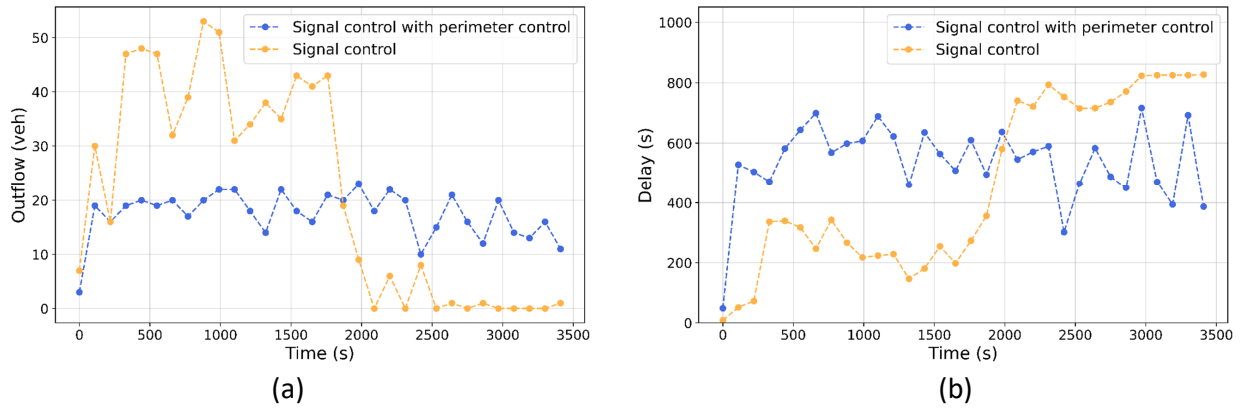


Figure S7. Comparison of outflow and delay evolution at intersection A with and without perimeter control. (a) Outflow evolution. (b) Delay evolution.

spatial density distribution in Figure S6b further demonstrates a shift toward lower density levels (green coloring), particularly within the area marked by the black dashed box. Moreover, perimeter control regulates inflows at boundary intersections, as evidenced by the outflow and delay evolution at intersection A in Figure S7 (blue lines), resulting in more stable release and enhanced traffic efficiency from exceeding network capacity.

Furthermore, vehicular-level control is implemented to provide microscopic management within the hierarchical control architecture. A case study considers a single vehicle traveling from a designated origin to a destination within the Xuancheng network along a high-volume corridor, represented by the straight-through path from intersection A in Figure S6a. Vehicle trajectories in Figure S8a indicate that the direct route suffers from severe congestion, preventing the vehicle from reaching its destination within 400 seconds. To mitigate the bottleneck, an alternative, less congested path with a longer distance is assigned. As illustrated in Figure S8b, the detour achieves a shorter travel time compared to the direct route, despite its longer distance.

In summary, the multidimensional evaluation validates that the proposed hierarchical control architecture enables effective coordination of control strategies across different operational levels, thereby improving traffic flow, alleviating congestion, and enhancing overall network efficiency.

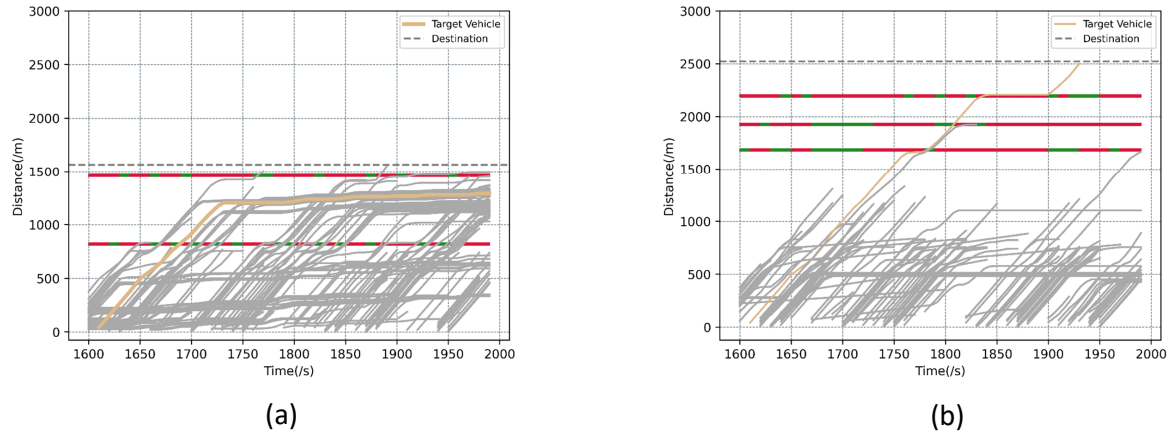


Figure S8. Comparison of vehicle trajectories before and after vehicle routing. **(a)** Direct route through congested corridor. **(b)** Detour route avoiding congested corridor.

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